

AHMED MOHAMED HASSAN MOHAMED KHALIFA

Mechatronics Engineer | Robotics | Embedded | .NET Developer

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PROFILE

Recent Mechatronics Engineering graduate from Misr University for Science and Technology, currently working as a Teaching Assistant and Production Engineer. Hands-on background in robotics, embedded systems, automation, CNC machines, PLC programming and production engineering. Also experienced in building software and full stack systems using C#, ASP.NET Core MVC, SQL Server, React, Node.js, ESP32 APIs and database-driven web applications.

TECHNICAL SKILLS

Software & Full Stack	C#, ASP.NET Core MVC, .NET 8, Entity Framework Core, SQL Server, SSMS, Identity, WinForms, DevExpress, REST APIs, React, Vite, Node.js, Express, MongoDB, Git, GitHub
Robotics & Embedded	ROS 1, ROS 2 Jazzy, C++, Arduino UNO/Mega/Nano, ESP32, sensors, motor drivers, LCD/I2C, serial communication, Proteus, EasyEDA, PCB design
Engineering Tools	MATLAB, Simulink, SolidWorks, AutoCAD, PLC, TIA Portal, CNC, G-code, Dialux, Excel, Word

EXPERIENCE & EDUCATION

EMPLOYMENT

Teaching Assistant | Misr University for Science and Technology

09/2024 - Present

Production Engineer | Giza Power Industry

02/2025 - Present

EDUCATION

Bachelor of Engineering

Mechatronics Engineering

Misr University for Science and Technology | Sep 2019 - Jul 2024

PROJECTS

Smart Power System Using Microgrid and AI A smart energy-management project using microgrid concepts, artificial intelligence, MATLAB and Simulink to improve power monitoring, analysis and control. AI MATLAB Simulink Microgrid	Restaurant Website A modern responsive restaurant website designed to display menu items, restaurant information, contact details and a clean user experience. HTML CSS JavaScript Web Development
Machine Vision Barcode and QR Code Reader A machine vision system designed to read barcodes and QR codes, save scanned data into a database and count each barcode for tracking, inventory and automation. Machine Vision Barcode QR Code Database	Automatic Bottle Filling System An automation project designed to fill bottles automatically using sensors, control logic and actuators to improve production speed and reduce manual work. Automation Sensors Control System Industrial Design
ASP.NET Core Store Website A full stack e-commerce store built with ASP.NET Core MVC, C#, Entity Framework Core, SQL Server and Identity, including products, cart, checkout, orders, admin dashboard, image upload, Excel import and email features. .NET 8 MVC EF Core SQL Server Identity	Supermarket Cashier Desktop App A C# desktop cashier system for product management, barcode workflow, sales, database integration, cancel button logic and report viewing using DevExpress. C# WinForms SQL Server DevExpress

Habit Mood ESP32 Sensor Dashboard A real-time IoT health dashboard using ESP32 sensor data, React/Vite frontend, Node.js/Express backend and MongoDB to display temperature, heart rate, SpO2, steps, status and charts. React Node.js MongoDB ESP32 API	Smart Solar Grass-Cutting Robot An autonomous solar robot designed for grass cutting using ROS 1, mapping and navigation, combining mobile robot control, environment mapping, path planning, solar energy and embedded motor control. ROS 1 Mapping Navigation Solar Robot Automation
Quad Magnetic Stirrer A hardware control system using Arduino Mega, four DC motors, two L298N drivers, LCD display, switches and potentiometers for speed/time control with countdown stopping logic. Arduino Mega L298N LCD Proteus PCB Design	ROS 2 Jazzy Robotics Projects ROS 2 C++ practice packages including publishers, subscribers, services, turtlesim motion, Arduino serial communication, sensor publishing and rqt_graph/rqt_plot debugging. ROS 2 C++ rclcpp Serial Turtlesim
4-DOF Robotic Arm Simulation A MATLAB/Simulink robotic arm project using DH parameters, forward/inverse kinematics, trajectory drawing and Simscape Multibody simulation concepts. MATLAB Simulink DH IK/FK	Automation & PLC Training Projects Automation practice based on PLC, TIA Portal, CNC, production engineering and industrial troubleshooting experience. PLC TIA Portal CNC Automation

COURSES & TRAINING

Automotive Training Course Engovation Aug 2024 - Sep 2024	Robot Operating System (ROS) Robotics Corner Jun 2023 - Mar 2024
TIA Portal Programming with S7-1200 & S7-1500 Arab Institute for Advanced Technology Jan 2023 - Feb 2023	Basics of Programming, Operation CNC Turning & Milling Machines Arab Institute for Advanced Technology Sep 2022
Flight Simulator Training EgyptAir Academy Aug 2022 - Sep 2022	Electronics Devices Assembly Workshop Arab Institute for Advanced Technology Sep 2021

STRENGTHS & LANGUAGES

Strengths: Adaptability, effective time management, creativity, leadership, teamwork, communication, ability to work under pressure, fast learning and computer skills.	Languages: Arabic, English, French
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